

Metamodel of the REA model level

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Abstract – This paper describes the gradual formation of the metamodel of the REA model level of the REA enterprise ontology. The paper aims to create a basic structure that allows creating valid models of the ontology. The first part of the paper discusses the identification of the elements used as a basis for the metamodel and their relationships. The second part describes the creation of the metamodel itself. This metamodel serves as the fundament for creating domain-specific modelling tool and can be used as a basis for an information system based on the REA enterprise ontology.

Keywords – REA ontology, metamodel, validation, identification, business process modelling

1. Introduction

The business processes must be constantly developed and improved to ensure the competitiveness of the company nowadays. It cannot be done without their modelling [1]. There are many business processes modelling formalism, but due to inadequate level of an abstraction and the use of general concepts they are not usable enough for business process modelling. Therefore companies usually use the expensive software created directly to their specific requirements.

The REA enterprise ontology is the concept for designing and a model creation of enterprise infrastructures. It is based on the resource ownership and its exchange [4]. The REA ontology does not use general concepts but specific. It increases the amount of represented data, while maintaining the simplicity of the model. The REA ontology model has an object-oriented structure and includes internal rules for verifying the consistency of the model, ensuring correctness of created links. At the same time models are simple and understandable for ordinary users who will work with it, but sufficiently precise for its automation [5].

The core of the REA ontology is the REA model level that describes individual business processes. This level shows particular changes in the value of resources. The value is based on the concepts of economic exchanges and conversions increasing the value of the company. In order to create the metamodel that will provide the basic validation of the REA model level, it is necessary to identify elements and their relations in the model. According

to [3] the metamodel is a model that is used for formal specification of other models. In the case of this paper the metamodel defines the structure of the language of the REA model level. This metamodel can serve as the cornerstone for applications that use the REA ontology, such as modelling tools, economic simulators, accounting systems and others.

The metamodel of the REA ontology's domain has been processed several times. One of the available metamodels was created for the modelling tool REA-DSL [7], another one was introduced for the definition of business rules [2]. The first mentioned is restricted to the operational level (base) of the REA model level and therefore it is not usable for tools in practice. The second metamodel has been modified to suit to a particular use (specification of business rules) and it is not consistent with the structure of the ontology as described in [4]. None of available metamodels of the REA ontology is general and therefore they cannot be used to create variety tools.

This article describes the gradual creation of the metamodel. First, basic elements of the model, their relations and domain rules are identified. Then based on the previous identification the metamodel is sequentially built. The resulting metamodel is sufficient to serve as a basis of a modelling tool and it can be further expanded and modified as required to reflect the manner of use.

2. Identification of key elements of the REA model level

The first step to build the metamodel is to identify individual elements of the modelled ontology. In this step it is necessary to specify all basic elements that form the base of the metamodel and their properties. The basic structure of the REA model level is divided into two groups: the operational level and the policy level.

The operational level includes concepts forming the name of the ontology - economic resources, economic events and economic agents. This level describes specific facts that have already occurred.

The policy level contains concepts describing what could, should, or should not happen. In this category concepts and semantic abstractions such as scheduling, grouping, contracts and others are included. Currently the REA ontology can cover a wide range of economic problems with a limited

number of concepts. Some of these concepts have limited use focused on solving problems on specific economic areas. For this reason it is not appropriate to deal with all concepts. The paper deals with basic and the most frequently used concepts of the REA model level. Created metamodel can be expanded by other concepts later.

Basic operational level of the REA model level contains 3 concepts [6]:

- *Economic resource* – Basic economical company resource that company wants to plan, monitor and control. Examples of economic resources are products, services, money, raw materials, tools, etc.
- *Economic agent* - An individual or some organization capable of having control over economic resources and able to receive or transmit further control of other agents. An example might be a customer, supplier, company, etc.
- *Economic event* – Represents either increment or decrement in the value of economic resource. This transformation could be realized immediately or in a certain time period. Examples could be work unit, using of services, renting, etc.

Selected policy level concepts included in the metamodel are:

- *Typification* - It is a homogeneous collection whose elements have the same characteristics defined by the type. Typification is an abstraction of a group of objects into a certain category forming bond "is a-kind-of" It is used to capture the description of the concept applied to a set of objects [5].
- *Group* - Is a structural element of the REA model level designed for creating heterogeneous collections or sets of REA entities. This is a special form of aggregation that forms a bond "is a-member-of" [5]. The group itself is not limited to operating level entities, but virtually any entity of the REA model level including other groups with which, however, forms intransitive dependence.
- *Commitment* - Promise or obligation to perform an economic event in a specific time. For this reason, parameters should contain the scheduled date and planned value. An example can be ordering goods, where one party agrees to provide chosen goods and the other party agrees to provide the required amount.
- *Contract/Schedule* - Is a set of obligations and rules that define conditions of performing the duality in the future and events performed if initial commitments are unfulfilled.

Entity type is divided by the object that aggregates into three elements: resource type, event type and agent type. Although the REA ontology allows creating types of policy level concepts (see [4]), they are almost not used in practice and therefore they were not included in the metamodel.

Among the fundamental concepts and semantic abstractions mentioned in the table the metamodel also includes elements providing M:N relationships. This relationship occurs where several inputs are connected to several outputs. Such a situation is usually solved by the node. In the case of the REA model level this relationship occurs mainly as a representation of the duality link (link between incremental and decrement events) and reciprocity link (link between incremental and decrement commitments). In the model, this node is shown as a rhomboid.

The link between elements in the metamodel allows the interconnection in the final model. By the definition of links the basic validation of the model is achieved. Model can only create links between such elements, among which is binding determined in the metamodel. This prevents the creation of incorrect links that should not exist in the model.

Based on domain rules and restrictions the cardinality of links is set. This allows determining the validation of domain rules on the metamodel level and therefore is not necessary to specify them in the validation framework.

The following list contains relationships and their restrictions, which are used to determine additional validation criteria. These criteria are applied during the creation of the metamodel as multiplicity constraints.

- *Stack flow (Inflow/ Outflow)* - Connects economic resource and economic event
 - There must be at least one economic event for every economic resource.
 - There must be at least one economic resource for every economic event.
- *Participate (Provide/ Receive)* – Connects economic agent and economic event
 - Every event requires the participation of two agents, one in the role of the recipient and the second as a provider.
 - Every agent must be connected with an incremental event by the *Receive* link and with a decrement event by the *Provide* link.
- *Duality* - Connects an incremental and a decrement economic event

- Every incremental event must be linked to at least one decrement event through Duality entity.
 - Every decrement event must be linked to at least one incremental event through Duality entity.
- *Typification* - Connects resource/event/agent type with corresponding concept of operational level
 - Every entity type must be linked with just one entity it represents.
- *Fulfillment* - Connects commitment and economic event
 - Every incremental commitment must be linked to incremental economic event.
 - Every decrement commitment must be linked to decrement event.
- *Specification* - Connects commitment and agent (or agent type)
 - Every commitment must specify the agent or the agent type who is responsible for it.
- *Reciprocity* - Connects incremental and decrement commitment
 - Every decrement commitment must be linked with at least one incremental commitment.
 - Every incremental commitment must be linked with at least one decrement commitment.
- *Clause* - Connects commitment with contract or schedule.
 - Every contract must contain decrement and incremental commitment.
 - Every commitment must be declared by some contract or schedule.
- *Reservation* - Connects commitment and economic resource or resource type
 - Every commitment has to reserve at least one type of resource.
- *Group* - Connects group with any entity of the model
 - It has no restrictions.

3. Creation of the metamodel

Creation of the metamodel is analogous as the formation of the model itself. Individual model elements are sequentially added to the metamodel. These elements are interconnected by specific links that correspond to possible connection in the model. Based on domain rules the multiplicity is defined for these links. By this way the behavior of the model, its elements and links is captured in the metamodel and the basic model validation is formed.

The base of the REA model level is the operational level containing three concepts: Resource, Event and Agent. An economic event is divided into incremental or decrement event.

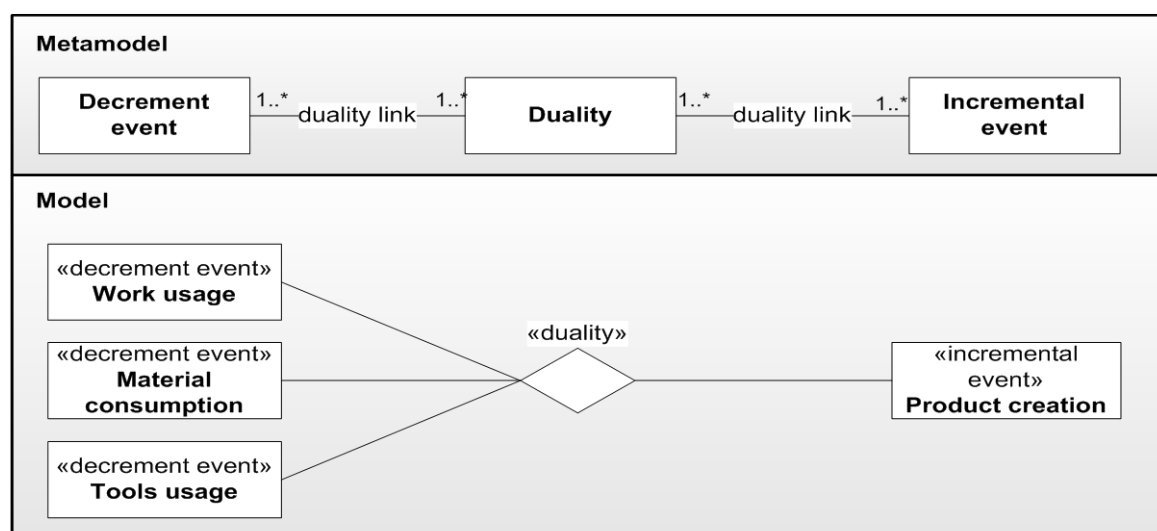


Figure 1. Realization of the duality link in the metamodel and its usage in the model

- *Party* - Connects contract/schedule and economic agent or agent type

Each contract/schedule must declare two parties (Agents or Agent types).

These events are instances of the parent concept Event that defines their internal structure.

Incremental and decrement events are mutually dual. This duality (relation between incremental and decrement events) has cardinality M: N, where one or more incremental events are associated with one

or more decrement. To be able to illustrate such a relationship in a model, it is necessary to use the connecting entity which performs the role of the node. Duality has its own entity in the metamodel and in the model it is represented by a rhomboid. Figure 1 shows the duality solution in the metamodel (upper part) and its usage in the model (bottom part). The example of the model contains three decrement events (Work usage, Material consumption and Tools usage) which are mutually dual with one incremental event (Product creation).

An economic event is linked to an economic resource. In case of a decrement event it is the *Outflow link* and in case of an incremental event it is the *Inflow link*. An economic agent forms two links with economic events - the *Provide link* and the *Receive link*. Because these links are applied independently on the type of the event (whether it is an incremental or decrement), it is possible to simplify the metamodel by connecting them directly to the Event, instead of its instances. Figure 2 shows the part of the metamodel containing concepts and relationships of the operational level. This metamodel allows creating basic models of this level.

also the *Commitment* is divided into incremental and decrement. Because the incremental commitment forms different links than decrement commitment, it is necessary to implement this division in the metamodel. It is necessary for the metamodel to include instances of the *Commitment - Incremental* and *Decrement commitment*. The link between these instances is *Reciprocity*. This relationship is analogous to the *Duality* between incremental and decrement events and as well as duality it is solved by the connecting entity. Any commitment independently on its type reserves an economic resource or a resource type. In both cases it creates a *Reservation link*. The link is the same because the metamodel must allow creating the same connection between different elements in the model. Between the commitment and the agent is a *Specification link*. Agent type can be used instead of an economic agent, but the link between it and commitment is the same. The relationship between commitment and the event is called *Fulfilment*. This relationship is type-dependent, so incremental commitment can connect to incremental events and vice versa. In special cases (such as planning) the *Fulfilment link* can connect a commitment with an event type.

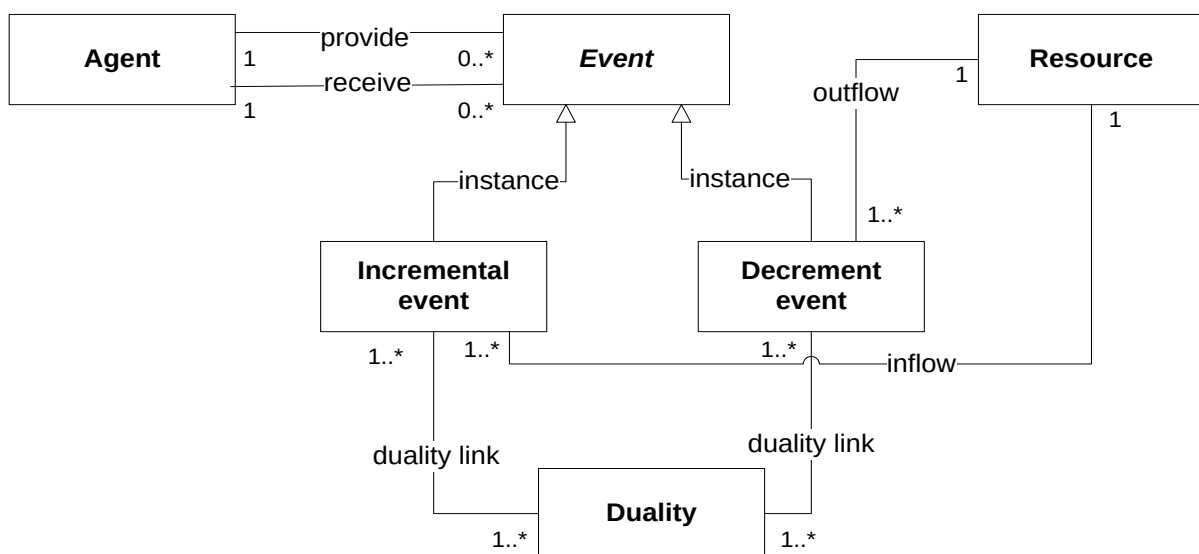


Figure 2. Metamodel of the operational level of the REA model level

Once the operational level is completed, the metamodel can be extended by concepts and semantic abstractions of the policy level. The most commonly used semantic abstraction is the *Typification*. It contains three elements specified by appropriate concept of the operational level - the *Agent type*, *Event type* and *Resource type*. Every *Type* element is attached to a proper operating level concept with which it forms the *Typification link*.

The most common use of typification is within the *Commitment* for making reservations. Like the *Event*

Another concept included in the metamodel is the *Contract*. Every contract in the model defines parties through the relation *Party*. This relationship must connect a contract with at least two agents or agent types. Minimum number of agents and agent types connected to the contract cannot be solved at the metamodel level by cardinality, because the metamodel cannot cover every combination of links *Contract-Agent* and *Contract-Agent type*. The last relationship of the contract is the *Clause link* between the *Contract* and the *Commitment*.

The last concept included in the metamodel is the *Schedule*. This is analogous to the *Contract*, but it has different cardinality constraints. Unlike the *Contract* it requires the participation of at least one agent or agent type using the *Party* link and the difference is also in the *Clause* link cardinality. It says that the *Commitment* in the model is not necessarily connected to the *Schedule*.

The second semantic abstraction in the metamodel is the *Group*. It can connect all elements in the model, including itself without restrictions. Because

the *Group* is heterogeneous collection, it is not limited to one particular kind of element, as in the case of *Typification*, but it can group any model elements.

Figure 3 shows the resulting metamodel of presented concepts and semantic abstractions of the REA model level. To maintain clarity of the figure, links of the semantic abstraction *Group* was omitted. Elements in the metamodel are coloured by the levels - elements of the operational level are shown in white, elements of the policy level are gray.

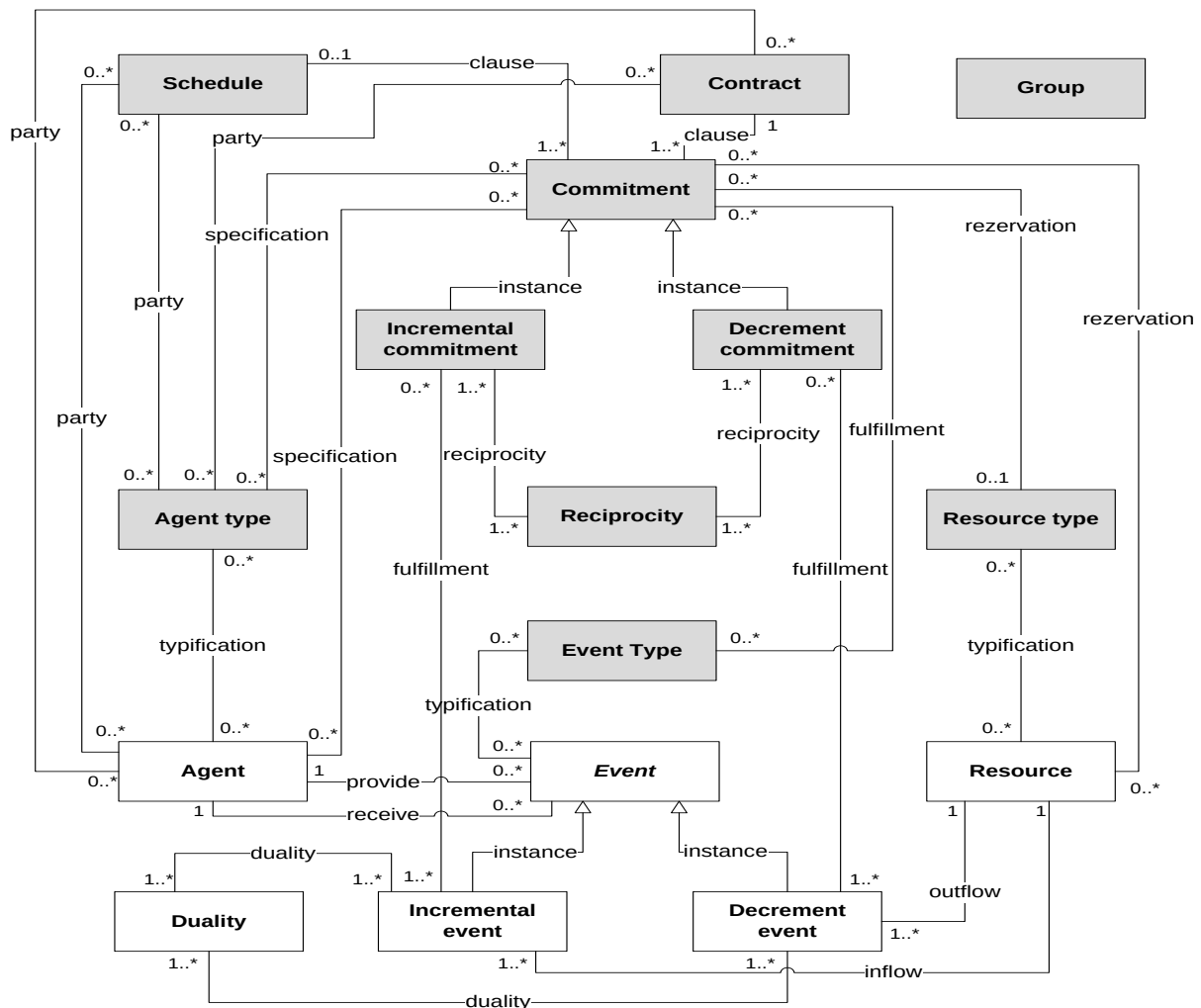


Figure 3. Metamodel of selected concepts and semantic abstractions of the REA model level

4. Conclusion

This paper introduced the procedure of creating the metamodel of the REA model level intended as crucial part of the modelling tool. In the individual chapters all steps leading to the resulting assembly were sequentially introduced. In the first part of the

paper key elements, their binding and domain rules were identified. The second part describes the sequentially creation of the metamodel itself. The result is the metamodel of the REA model level (figure 3), which is able to provide basic validation of the model and can serve as a cornerstone for other applications.

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